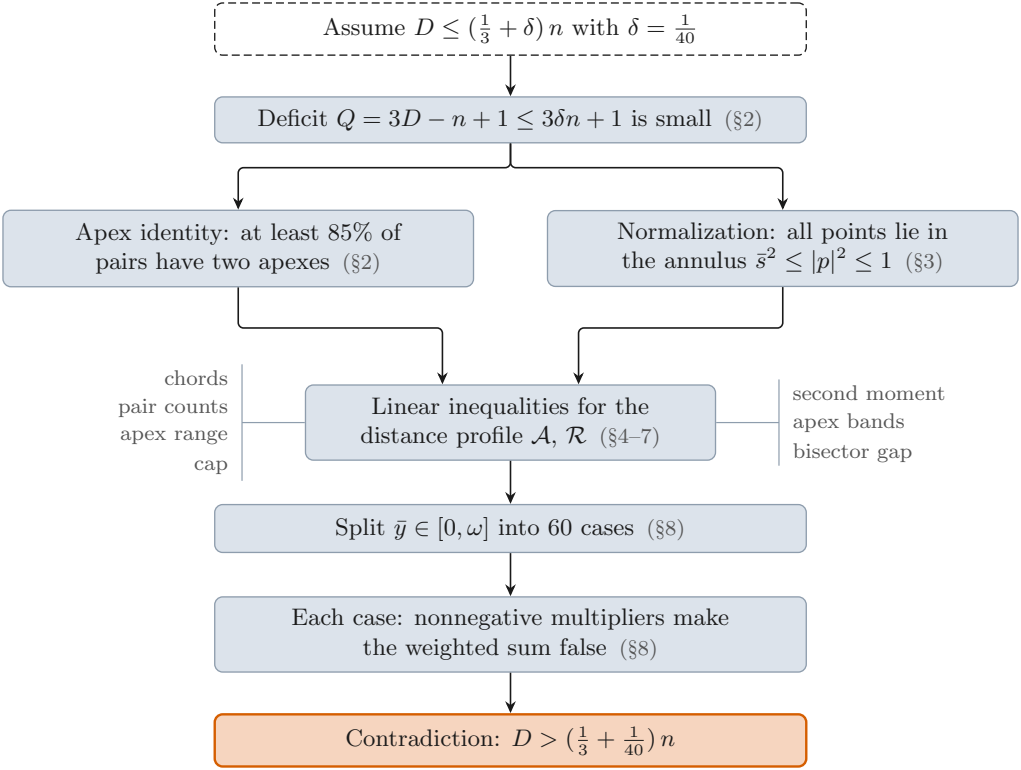

Assume $D \le (\frac{1}{3} + \delta)\, n$ with $\delta = \frac{1}{40}$
Deficit $Q = 3D - n + 1 \le 3\delta n + 1$ is small (§2)
Apex identity: at least 85% of pairs have two apexes (§2)
Normalization: all points lie in the annulus $\bar{s}^2 \le |p|^2 \le 1$ (§3)
chords
pair counts
apex range
cap
Linear inequalities for the distance profile $\mathcal{A}$, $\mathcal{R}$ (§4–7)
second moment
apex bands
bisector gap
Split $\bar{y} \in [0, \omega]$ into 60 cases (§8)
Each case: nonnegative multipliers make the weighted sum false (§8)
Contradiction: $D > (\frac{1}{3} + \frac{1}{40})\, n$